

11. Chronic Kidney Disease and Risk Management

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Chronic Kidney Disease and Risk Management: Standards of Care in Diabetes—2026

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For prevention and management of diabetes complications in children and adolescents, please refer to section 14, “Children and Adolescents.”

EPIDEMIOLOGY OF DIABETES AND CHRONIC KIDNEY DISEASE

Chronic kidney disease (CKD) is diagnosed by the persistent elevation of urinary albumin excretion (albuminuria), estimated glomerular filtration rate (eGFR) < 60 mL/min/1.73 m², or other manifestations of kidney damage (1). In this section, the focus is on CKD attributed to diabetes in adults, which occurs in 20–40% of people with diabetes (1–4). CKD in people with diabetes typically develops after a duration of 10 years in type 1 diabetes (the most common presentation is 5–15 years after the diagnosis of type 1 diabetes) but may be present at diagnosis of type 2 diabetes. CKD can progress to kidney failure requiring dialysis or kidney transplantation and is the leading cause of kidney failure in the U.S. (5). In addition, among people with type 1 or type 2 diabetes, the presence of CKD markedly increases cardiovascular risk and health care costs (6). The incidence of CKD in people with diabetes has decreased over the last decade but remains high in the U.S. (3) and around the world (7). For details on the management of CKD in children and adolescents with diabetes, please see section 14, “Children and Adolescents.”

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ASSESSMENT OF ALBUMINURIA AND ESTIMATED GLOMERULAR FILTRATION RATE

Recommendations

11.1a Assess kidney function with random urine albumin-to-creatinine ratio (UACR) and estimated glomerular filtration rate (eGFR) at least annually in people with type 1 diabetes with duration of ≥ 5 years and in all people with type 2 diabetes regardless of treatment. B

11.1b In people with chronic kidney disease (CKD), monitor urinary albumin (e.g., spot UACR) and eGFR 1–4 times per year depending on the stage of kidney disease (Fig. 11.1). B

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Acute kidney injury (AKI) is defined as a rapid decline in kidney function, typically within a short period (hours to days), characterized by increase in serum creatinine and/or decrease in urine output (26). People with diabetes

are at higher risk of AKI than those without diabetes (27). The rates of hospitalizations for AKI requiring dialysis are five times higher in adults with diabetes compared with adults without diabetes (risk ratio 5.0; 95% CI 4.8–5.1) (28). Other risk factors for AKI include preexisting CKD, the use of medications that cause kidney injury (e.g., nonsteroidal anti-inflammatory drugs), certain intravenous dyes (e.g., iodinated radiocontrast agents), and the use of medications that alter kidney blood flow and intrakidney hemodynamics.

DIAGNOSIS OF CHRONIC KIDNEY DISEASE IN PEOPLE WITH DIABETES

CKD in people with diabetes is usually a clinical diagnosis made based on the presence of albuminuria and/or reduced eGFR in the absence of signs or symptoms of other primary causes of kidney damage. The typical presentation of CKD in people with diabetes is considered to include long-standing duration of diabetes, retinopathy, albuminuria without gross hematuria, and gradually progressive loss of eGFR. However, signs of CKD may be present at diagnosis or without retinopathy in type 2 diabetes. Reduced eGFR without albuminuria has been frequently reported in type 1 and type 2 diabetes and is becoming more common over time as the prevalence of diabetes increases in the U.S. (2,3,18,20,21).

An active urinary sediment (containing red or white blood cells or cellular casts), rapidly increasing albuminuria or total proteinuria, the presence of nephrotic syndrome, rapidly decreasing eGFR, or the absence of retinopathy (particularly in type 1 diabetes) suggests alternative or additional causes of kidney disease. For individuals with these features, referral to a nephrologist for further diagnosis, including the possibility of kidney biopsy, should be considered. It is rare for people with type 1 diabetes to develop kidney disease without retinopathy. In type 2 diabetes, retinopathy is only moderately sensitive and specific for CKD caused by diabetes, as confirmed by kidney biopsy (22).

STAGING OF CHRONIC KIDNEY DISEASE

CKD is defined in stages by level of eGFR and albuminuria, with stage G1 and stage G2 CKD defined by evidence of high albuminuria with eGFR ≥ 90 mL/min/1.73 m² and eGFR ≥ 60 but < 90 mL/min/1.73 m², respectively. Stages G3-G5 CKD are defined by progressively lower ranges of eGFR (23). Stage G5 is kidney failure (Fig. 11.1). At any eGFR, the degree of albuminuria is associated with risk of cardiovascular disease (CVD), CKD progression, and mortality (6). Therefore, there is an additional subclassification by level of urine albumin (Fig. 11.1) (1).

Thus, based on the current classification system, both eGFR and albuminuria must be quantified to guide treatment decisions. Quantification of eGFR levels is essential for modifications of medication dosages or restrictions of use (Fig. 11.1) (23,24), and the degree of albuminuria should influence the choice of antihypertensive medications (see section 10, “Cardiovascular Disease and Risk Management”) or glucose-lowering medications (see below). Observed history of eGFR loss (which is also associated with risk of CKD progression and other adverse health outcomes) and cause of kidney damage (including possible causes other than diabetes) may also affect these decisions (25).

Table 11.1—Reasons to consider nondiabetic kidney diseases in a person with chronic kidney disease and diabetes

- Type 1 diabetes duration ≥ 5 years or no retinopathy
- Active urine sediment (e.g., containing red blood cells or cellular casts)
- Chronically well-managed blood glucose
- Rapidly declining eGFR
- Rapidly increasing or very high UACR (> 300 mg/g) or urine protein/creatinine level (> 500 mg/g)

Information adapted from Liang et al. (133). eGFR, estimated glomerular filtration rate; UACR, urine albumin-to-creatinine ratio.

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cartilage glycoprotein 40) did not increase in individuals assigned to intensive blood pressure control in the Systolic Blood Pressure Intervention Trial (SPRINT) trial despite decrease in eGFR (37).

Accordingly, ACE inhibitors and ARBs should not be discontinued for increases in serum creatinine ($\geq 30\%$) in the absence of volume depletion. Similar rises in creatinine are seen at the initiation of SGLT2

inhibitors and glucagon-like peptide 1 receptor agonists (GLP-1 RA) due to hemodynamic changes in the tubuloglomerular feedback, and the medication should not be discontinued.

SURVEILLANCE

| Complication | Physical and laboratory evaluation | | ----- |
----- | | Blood pressure >130/80 mmHg | Blood pressure, weight, BMI | | Volume overload | History, physical examination, weight | | Electrolyte abnormalities | Serum electrolytes | | Metabolic acidosis | Serum electrolytes | | Anemia | Hemoglobin; iron, iron saturation, ferritin testing if indicated | | Metabolic bone disease | Serum calcium, phosphate, PTH, vitamin 25(OH)D |

Complications of chronic kidney disease (CKD) generally become prevalent when estimated glomerular filtration rate falls below 60 mL/min/1.73 m² (stage G3 CKD or greater) and become more common and severe as CKD progresses. Evaluation of elevated blood pressure and volume overload should occur at every clinical contact possible; laboratory evaluations are generally indicated every 6–12 months for stage G3 CKD, every 3–5 months for stage G4 CKD, and every 1–3 months for stage G5 CKD, or as indicated to evaluate symptoms or changes in therapy. 25(OH)D, 25-hydroxyvitamin D; PTH, parathyroid hormone.

Recommendation

11.2 Aim to reduce urinary albumin by ≥30% in people with CKD and albuminuria ≥300 mg/g to slow CKD progression. B and achievement of blood pressure goals. Early changes in kidney function may be detected by increases in albuminuria before changes in eGFR (41), and this also significantly affects cardiovascular risk.

Both albuminuria and eGFR should be monitored annually to enable timely diagnosis of CKD, monitor progression of CKD, detect superimposed kidney diseases including AKI, assess risk of CKD complications, dose medications appropriately, and determine whether nephrology referral is needed. Among people with existing kidney disease, albuminuria and eGFR may change due to progression of CKD, development of a separate superimposed cause of kidney disease, AKI, or other effects of medications, as noted above. Serum potassium should also be monitored in individuals treated with diuretics because these medications can cause hypokalemia, which is associated with cardiovascular risk and mortality (38–40). Individuals with eGFR <60 mL/min/1.73 m² receiving ACE inhibitors, ARBs, or mineralocorticoid receptor antagonists (MRAs) should have serum potassium measured periodically to assess for hyperkalemia. Additionally, people with this lower range of eGFR should have their medication dosing verified, their exposure to nephrotoxins (e.g., nonsteroidal anti-inflammatory drugs and iodinated contrast) should be minimized, and they should be evaluated for potential CKD complications (Table 11.2).

Prevention

The only proven primary prevention interventions for CKD in people with diabetes are blood glucose (A1C goal of 7%) and blood pressure management (<130/80 mmHg). There is no evidence that RAS inhibitors or any other interventions prevent the development of CKD in the absence of hypertension or albuminuria. Thus, the American Diabetes.

Interventions that lower albuminuria • Blood glucose management • Blood pressure management • Treatment with ACE inhibitors or ARBs • Smoking cessation • Weight loss • Changes in eating patterns (decreased salt intake and/or protein intake) • Physical activity • Treatment with SGLT2 inhibitors, MRAs, or GLP-1 RAs

ARB, angiotensin receptor blocker; GLP-1 RA, glucagon-like peptide 1 receptor agonist; MRA, mineralocorticoid receptor antagonist; SGLT2, sodium–glucose cotransporter 2.

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Association does not recommend routine use of these medications solely for the purpose of prevention of the development of CKD. In 2023, the Glycemia Reduction Approaches in Diabetes: A Comparative Effectiveness Study (GRADE) was published. This large prospective study compared liraglutide, sitagliptin, glimepiride, and insulin glargine with respect to achieving and maintaining A1C goals in people with type 2 diabetes treated with

metformin monotherapy; kidney and cardiovascular end points were examined as secondary outcomes. A total of 5,047 participants were enrolled from July 2013 to August 2017 and were followed for an average of 5 years. Almost all participants did not have signs of kidney disease at the time of enrollment. No differences between the examined medications were observed for kidney-protective effects among these medications for prevention. Of note, SGLT2 inhibitors were not included in the study, as these medications were not routinely available at the time the study started.

INTERVENTIONS

Nutrition

Recommendation

11.3 For people with CKD stage G3 or higher, protein intake should be 0.8 g/kg body weight per day, as for the general population. A For individuals on dialysis, protein intake of 1.0–1.2 g/kg/day should be considered, since protein energy wasting is a major problem for some individuals on dialysis. B

For people with stage G3–G5 non–dialysis-dependent CKD, protein intake should be ~0.8 g/kg body weight per day (the recommended daily allowance). Compared with higher levels of protein intake, this level slowed GFR decline with evidence of a greater effect over time. Higher levels of protein intake (>20% of daily calories from protein or >1.3 g/kg/day) have been associated with increased albuminuria, more rapid kidney function loss, and CVD mortality and therefore should be avoided. In people with CKD and diabetes, consuming protein below the recommended daily allowance of 0.8 g/kg/day is not recommended, because it does not alter blood glucose levels, cardiovascular risk measures, or the risk of CKD progression.

Blood Pressure and Use of ACE Inhibitors and Angiotensin Receptor Blockers

Recommendations

11.5 If it can be safely attained, the on-treatment blood pressure goal for people with CKD is <130/80 mmHg; a systolic blood pressure goal <120 mmHg and/or reduction in blood pressure variability should be encouraged. A

11.6a In nonpregnant people with diabetes and hypertension, either an ACE inhibitor or an angiotensin receptor blocker (ARB) is recommended for those with moderately increased albuminuria (UACR 30–299 mg/g creatinine) B and is strongly recommended for those with severely increased albuminuria (UACR ≥300 mg/g creatinine) and/or eGFR <60 mL/min/1.73 m² to maximally tolerated dose to prevent the progression of kidney disease and reduce cardiovascular events. A If one class is not tolerated, the other should be substituted. B

11.6b Monitor for drop in eGFR and increase in serum potassium levels at initiation and periodically as clinically appropriate when ACE inhibitors, ARBs, and mineralocorticoid receptor antagonists (MRAs) are used. B Monitor for hypokalemia when diuretics are used.

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at routine visits and 7–14 days after initiation or after a dose change and periodically as clinically appropriate. B

11.6c An ACE inhibitor or an ARB is not recommended for the primary prevention of CKD in people with diabetes who have normal blood pressure, normal UACR (<30 mg/g creatinine), and normal eGFR.

11.6d Continue renin-angiotensin system blockade for mild to moderate increases in serum creatinine (≤30%) in individuals who have no signs of extracellular fluid volume depletion. A

Blood Pressure Control Target in Diabetes (BROAD), an open-label trial that randomized 12,821 individuals with diabetes and elevated CVD risk to an intensive SBP goal of ≤120 mmHg or standard treatment that targeted an SBP of ≤140 mmHg for up to 5 years, noted a 21% reduction in nonfatal stroke, nonfatal myocardial infarction, heart failure, or cardiovascular death in the intensive SBP group (82). Incident albuminuria was also reduced in the intensive-treatment group compared with the standard-treatment group (11.29 vs. 13.84 events

per 100 person-years) (HR 0.87; 95% CI 0.77–0.97). There were similar incidences of CKD progression and CKD development.

Among individuals with diabetes and/or CKD receiving an ACE inhibitor or ARB therapy, SGLT2 inhibitor initiation was associated with a lower risk of hyperkalemia (HR 0.89 [95% CI 0.82–0.96]) and ACE inhibitor or ARB therapy discontinuation compared with nonusers (36% vs. 45%; $P = 0.001$) (88). In a joint analysis of two randomized, double-blind, placebo-controlled, event-driven trials of SGLT2 inhibitors in individuals with albuminuric CKD, the relative effect on ACE inhibitor or ARB blockade discontinuation was more pronounced among individuals with baseline UACR $\geq 1,000$ mg/g (pooled HR 0.77; 95% CI 0.63–0.94; P -heterogeneity = 0.009) (89).

Two recent large retrospective analyses provide additional support for the use of ACE inhibitors and ARBs at the maximum tolerated dose in individuals with CKD. Ku et al. (90) reviewed 17 trials that included 11,800 individuals with CKD (defined as eGFR ≤ 60 mL/min/1.73 m²); 82% had diabetes. The authors reported that a $\geq 13\%$ decline in eGFR over a 3-month period or a $\geq 21\%$ decline in a 1-month period was associated with better long-term kidney outcomes. Hattori et al. (91) evaluated 6,065 participants between 2005 and 2021 (~40% had diabetes) with eGFR ranging from 10 to 60 mL/min/1.73 m² who had ACE inhibitors or ARBs stopped (usually due to hyperkalemia or AKI) and found that those who restarted the ACE inhibitor or ARB had better long-term kidney outcomes and lower mortality (there was no significant difference in hyperkalemia in those who restarted ACE inhibitors or ARBs). There is also an accompanying editorial that details the strengths and weaknesses of the studies (92).

In the absence of kidney disease, ACE inhibitors or ARBs are useful to manage blood pressure but have not proven superior to alternative classes of antihypertensive therapy, including thiazide-like diuretics and dihydropyridine calcium channel blockers (93). In a trial of people with type 2

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Diabetes and normal urinary albumin excretion, an ARB reduced or suppressed the development of albuminuria but increased the rate of cardiovascular events (94). In a trial of people with type 1 diabetes exhibiting neither albuminuria nor hypertension, ACE inhibitors or ARBs did not prevent the development of glomerulopathy assessed by kidney biopsy (79). This was further supported by a similar trial in people with type 2 diabetes (80).

Two clinical trials studied the combinations of ACE inhibitors and ARBs and found no benefits on CVD or CKD, and the medication combination had higher adverse event rates (hyperkalemia and/or AKI) (95,96). Therefore, the combined use of ACE inhibitors and ARBs should be avoided.

Direct Kidney Effects of Glucose-Lowering Medications

Some glucose-lowering medications also have effects on the kidney that are direct, i.e., not mediated through glycemia. For example, SGLT2 inhibitors reduce kidney tubular glucose reabsorption, weight, systemic blood pressure, intraglomerular pressure, and albuminuria and slow GFR loss through mechanisms that appear independent of glycemia (31,97–101). Moreover, recent data support the notion that SGLT2 inhibitors reduce oxidative stress in the kidney by $>50\%$ and blunt increases in angiotensinogen as well as reduce NLRP3 inflammasome activity (100,102,103). GLP-1 RAs have also been shown to improve kidney outcomes (104–109). Nonmetabolic mechanisms by which GLP-1 RAs are believed to protect the kidney include anti-inflammatory.

Recommendations

- 11.7a For people with type 2 diabetes and CKD, use of a sodium–glucose co-transporter 2 (SGLT2) inhibitor with demonstrated benefit to reduce CKD progression and cardiovascular events is recommended. SGLT2 inhibitors should be initiated in individuals with eGFR ≥ 20 mL/min/1.73 m² but can safely continue until kidney failure. - 11.7b To reduce kidney disease progression and cardiovascular risk in people with type 2 diabetes and CKD, a glucagon-like peptide 1 agonist with demonstrated benefit in this population is recommended.

Lifestyle

| Healthy eating | Physical activity | Smoking cessation | Weight management | Regular risk factor reassessment (ever months) | | ----- | ----- | ----- | ----- | ----- |

RAS Therapy

ACEi or ARB (at maximal tolerated doses) should be first-line therapy for hypertension when albuminuria is present. Otherwise, dihydropyridine calcium channel blocker or diuretic can also be considered; all three classes are often needed to attain BP targets.

Additional Therapy

Add other agents with evidence of CVD and/or kidney benefit per individualized treatment goals and preferences (e.g., if on SGLT2i, and/or finerenone to further mitigate heart and kidney risk).

Regular reassessment of glycemia, albuminuria, BP, CVD risk, and lipids.

Figure 11.2

Holistic approach for improving outcomes in people with diabetes and CKD. Icons presented indicate the following benefits: BP cuff, BP lowering; glucose meter, glucose lowering; heart, cardioprotection; kidney, kidney protection; scale, weight management. eGFR is presented in units of mL/min/1.73 m².

Finerenone is currently the only nsMRA with proven clinical kidney and cardiovascular benefits. ACEi, angiotensin-converting enzyme inhibitor; ACR, albumin-to-creatinine ratio; ARB, angiotensin receptor blocker; ASCVD, atherosclerotic cardiovascular disease; BP, blood pressure; CCB, calcium channel blocker; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; GLP-1 RA, glucagon-like peptide 1 receptor agonist; HTN, hypertension; MRA, mineralocorticoid receptor antagonist; nsMRA, nonsteroidal mineralocorticoid receptor antagonist; PCSK9i, proprotein convertase subtilisin/kexin type 9 inhibitor; RAS, renin-angiotensin system; SGLT2i, sodium-glucose cotransporter 2 inhibitor; T1D, type 1 diabetes; T2D, type 2 diabetes.

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antioxidative, and immunomodulatory (117). GLP-1 RAs are suggested for cardio-vascular risk reduction if such risk is a pre-dominant problem, as they reduce risks of CVD events and slow progression of CKD (109,118–122).

Selection of Glucose-Lowering Medications for People With Chronic Kidney Disease

For people with type 2 diabetes and established CKD, special considerations for the selection of glucose-lowering medications include limitations to available medications when eGFR is diminished (such as increased risk for hypoglycemia) and a desire to mitigate risks of CKD progression, CVD, and hypoglycemia (112,113). Medication dosing may require modification with eGFR <60 mL/min/1.73 m² (1). Figure 11.2 shows the algorithm for medications in people with diabetes and CKD.

The U.S. Food and Drug Administration (FDA) revised its guidance for the use of metformin in CKD in 2016 (114), recommending use of eGFR instead of serum creatinine to guide treatment and expanding the pool of people with kidney disease for whom metformin treatment should be considered. The revised FDA guidance states that 1) metformin is contraindicated in individuals with an eGFR <30 mL/min/1.73 m², 2) eGFR should be monitored while taking metformin, 3) the benefits and risks of continuing treatment should be reassessed when eGFR falls to <45 mL/min/1.73 m² (115,116), 4) metformin should not be initiated for individuals with an eGFR <45 mL/min/1.73 m², and 5) metformin should be temporarily discontinued at the time of or before iodinated contrast imaging procedures in individuals with eGFR 30–60 mL/min/1.73 m².

Several studies have shown cardiovascular and kidney protection from SGLT2 inhibitors and GLP-1 RAs. Selection of which glucose-lowering medications to use should be based on the usual criteria of an individual's

risks (cardiovascular and kidney in addition to glucose management) as well as considerations of effects on weight, other adverse effects, other key comorbidities, individual preferences, and cost.

SGLT2 inhibitors are recommended for people with $\text{eGFR} \geq 20 \text{ mL/min/1.73 m}^2$ and type 2 diabetes, as they slow CKD progression and reduce heart failure risk independent of their effects on glucose.

A number of large cardiovascular outcomes trials in people with type 2 diabetes at high risk for CVD or with existing CVD examined kidney effects as secondary outcomes. These trials include EMPA-REG OUTCOME [BI 10773 (Empagliflozin) Cardiovascular Outcome Event Trial in Type 2 Diabetes Mellitus Patients], CANVAS (Canagliflozin Cardiovascular Assessment Study), LEADER (Liraglutide Effect and Action in Diabetes: Evaluation of Cardiovascular Outcome Results), REWIND (Researching Cardiovascular Events With a Weekly Incretin in Diabetes), and SUSTAIN-6 (Trial to Evaluate Cardiovascular and Other Long-term Outcomes With Semaglutide in Subjects With Type 2 Diabetes) (99,104,107,123).

Specifically, compared with placebo, empagliflozin reduced the risk of incident or worsening nephropathy (a composite of progression to $\text{UACR} > 300 \text{ mg/g}$ creatinine, doubling of serum creatinine, kidney failure, or death from kidney failure) by 39% and the risk of doubling of serum creatinine accompanied by $\text{eGFR} \leq 45 \text{ mL/min/1.73 m}^2$ by 44%; canagliflozin reduced the risk of progression of albuminuria by 27% and the risk of reduction in eGFR , kidney failure, or death from kidney failure by 40%; liraglutide reduced the risk of new or worsening nephropathy by 22%; dulaglutide reduced the composite kidney outcome by 25%; and semaglutide reduced the risk of new or worsening nephropathy by 36% (each $P < 0.01$).

These analyses were limited by evaluation of study populations not selected primarily for CKD and examination of kidney effects as secondary outcomes. Three large clinical trials of SGLT2 inhibitors have focused on people with CKD and assessment of primary kidney outcomes. Canagliflozin and Renal Events in Diabetes with Established Nephropathy Clinical Evaluation (CREDENCE), a placebo-controlled trial of canagliflozin among 4,401 adults with type 2 diabetes, $\text{UACR} \geq 300\text{--}5,000 \text{ mg/g}$, had a mean eGFR at baseline of $43.1 \pm 12.4 \text{ mL/min/1.73 m}^2$ (range $25\text{--}75 \text{ mL/min/1.73 m}^2$) and a median UACR of 949 mg/g (range $200\text{--}5,000 \text{ mg/g}$). There was a significant benefit by dapagliflozin for the primary end point (HR 0.61 [95% CI).

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0.51–0.72]; $P < 0.001$) (127). The HR for disease defined by level of eGFR and/or individuals at high risk of CKD progression by level of albuminuria (of note, all participants had an albuminuria level of at least 100 mg/g). The primary outcome was defined as the first major kidney disease event (onset of $>50\%$ in eGFR , onset of persistent eGFR of $\leq 15 \text{ mL/min/1.73 m}^2$, initiation of dialysis or transplant, kidney death, and cardiovascular death). The study was stopped early due to reaching a prespecified outcome. There was a 24% lower HR for those taking semaglutide compared with the placebo group. Of note, cardiovascular deaths comprised about 38% of the events. When the cardiovascular deaths are removed from the analysis, the HR for kidney-specific events was 21% lower in those taking semaglutide. Thus, this kidney outcome study supports a beneficial effect of semaglutide in slowing decline in kidney function as well as being cardioprotective in people with CKD and type 2 diabetes. Of note, the participants who took semaglutide had lower A1C, lower blood pressure, and more weight loss—all of which are beneficial for slowing decline in kidney function and reducing cardiovascular adverse events. Whether this beneficial combination of effects was the primary cause for the kidney-protective outcomes or whether there is a unique kidney-protective effect of semaglutide remains to be determined.

Of note, while the glucose-lowering effects of SGLT2 inhibitors are blunted with $\text{eGFR} \leq 45 \text{ mL/min/1.73 m}^2$, the kidney and cardiovascular benefits were still seen at eGFR levels as low as $20 \text{ mL/min/1.73 m}^2$ even with no significant change in glucose (4,29,31,107,123,127,129,130). Most participants with CKD in these trials also had diagnosed atherosclerotic cardiovascular disease (ASCVD) at baseline, although ~28% of CANVAS participants with CKD did not have diagnosed ASCVD (32). Cardiovascular and kidney events are reduced with SGLT2 inhibitor use in individuals with severe CKD independent of glucose-lowering effects (126,131). The six-year kidney benefits of SGLT2 inhibitors are comparable between empagliflozin and dapagliflozin for AKI, albuminuria, and CKD progression (132).

The FLOW study demonstrated that the GLP-1 RA semaglutide had kidney-protective effects in people with type 2 diabetes and CKD (109). The study enrolled 3,533 participants with significant kidney disease. SGLT2 inhibitors are recommended for people with type 2 diabetes and CKD, and the selection of specific agents may depend on comorbidity and CKD stage. Additional clinical trials focusing on CKD and cardiovascular outcomes in people with CKD are ongoing and will be reported in the next few years.

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Outcomes of Mineralocorticoid Receptor Antagonists in Chronic Kidney Disease

Recommendation

11.8 To reduce CKD progression and cardiovascular events in people with CKD and albuminuria, a nonsteroidal MRA that has been shown to be effective in clinical trials is recommended (if eGFR is ≥ 25 mL/min/1.73 m²). Potassium levels should be monitored 1 month after initiation.

The mean age of participants was 65.6 years, and 30% were female. The mean eGFR was 44.3 mL/min/1.73 m², and the mean albuminuria was 852 mg/g (interquartile range 446–1,634 mg/g). The primary end point was reduced with finerenone compared with placebo (HR 0.82 [95% CI 0.73–0.93]; $P = 0.001$), as was the key secondary composite of cardiovascular outcomes (HR 0.86 [95% CI 0.75–0.99]; $P = 0.03$). Hyperkalemia resulted in 2.3% discontinuation in the study group compared with 0.9% in the placebo group. However, the study was completed, and there were no deaths related to hyperkalemia. Of note, 4.5% of the total group were being treated with SGLT2 inhibitors.

The FIDELITY prespecified pooled efficacy and safety analysis incorporated individuals from both the FIGARO-DKD and FIDELIO-DKD trials ($n = 13,171$) to allow for evaluation across the spectrum of severity of CKD, since the populations were different (with a slight overlap) and the study designs were similar (145). The analysis showed a 14% reduction in composite cardiovascular death, nonfatal myocardial infarction, nonfatal stroke, and hospitalization for heart failure for finerenone vs. placebo (12.7% vs. 14.4%; HR 0.86 [95% CI 0.78–0.95]; $P = 0.002$).

It also demonstrated a 23% reduction in the composite kidney outcome, consisting of sustained $\geq 57\%$ decrease in eGFR from baseline over ≥ 4 weeks, or kidney death, for finerenone versus placebo (5.5% vs. 7.1%; HR 0.77 [95% CI 0.67–0.88]; $P = 0.001$).

The pooled FIDELITY trial analysis confirms and strengthens the positive cardiovascular and kidney outcomes with finerenone across the spectrum of CKD, irrespective of baseline ASCVD history (with the exclusion of those with heart failure with reduced ejection fraction).

Combination Therapy Considerations to Optimize Kidney and Cardiovascular Risk Reduction

Recommendation

11.9 Simultaneous initiation of an SGLT2 inhibitor and a nonsteroidal MRA (finerenone) can be considered in adults with type 2 diabetes and UACR ≥ 100 mg/g with eGFR 30–90 mL/min/1.73 m² on a renin-angiotensin system inhibitor due to evidence of safety and beneficial effects on albuminuria.

All of the recent kidney and cardiovascular outcomes studies for individuals with diabetes and CKD were completed in the setting of maximally tolerated RAS blockade. Thus, all individuals should have their treatment for RAS blockade adjusted to the maximally tolerated dose, even when considering combination therapy with other treatments. Although there have been no studies directly comparing SGLT2 inhibitors, GLP-1 RAs, or MRAs, the

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Health care professionals should use their best judgement, considering the individuals' comorbidities and preferences, as to which medication to prescribe initially (SGLT2 inhibitor or GLP-1 RA) and whether combination therapy is warranted. As noted, all of these studies included participants taking either an ACE inhibitor or an ARB, often at maximally tolerated doses. In addition, while current evidence of kidney benefit is limited to CKD in type 2 diabetes, multiple studies are in progress to determine if these therapies also improve kidney outcomes in people with type 1 diabetes (149,150).

Treatment for Severe Chronic Kidney Disease and Kidney Failure

Recommendation

11.11a Individuals with eGFR ≥ 20 mL/min/1.73 m² and not on dialysis can be safely continued on SGLT2 inhibitors to reduce the risk of CKD progression B and for cardiovascular benefits. C

11.11b Individuals on dialysis can be safely initiated or continued on GLP-1–based therapy that is not dependent on kidney clearance to reduce cardiovascular risk and mortality. C

SGLT2 inhibitors have non-kidney effects in multiple organs, including the cardiovascular system. For individuals who tolerate the medication, therapy should be continued unless kidney failure develops. A post hoc analysis of the DAPA-CKD trial demonstrated that in individuals with stage G4 CKD and albuminuria, the effects of dapagliflozin were consistent with those observed in the trial overall, with no evidence of increased risks (133). However, we will learn more about the safety and efficacy in kidney failure when results of the ongoing Renal Lifecycle trial are published. This study aims to investigate the effects of an SGLT2 inhibitor compared with placebo on the incidence of kidney failure, heart failure, mortality, and safety in individuals with an eGFR ≤ 25 mL/min/1.73 m² or kidney failure (153).

Small randomized controlled trials have demonstrated that GLP-1–based therapy has beneficial effects in body weight and glycemic management in individuals on dialysis. Idorn et al. (154) demonstrated that GLP-1 levels are increased in individuals on dialysis compared with individuals in the control group not on dialysis, and individuals on dialysis were more likely to experience gastrointestinal symptoms. In individuals receiving peritoneal dialysis, Hiramatsu et al. found over a 12-month period that left ventricular mass index decreased and left ventricular ejection fraction increased.

Vanek et al. (156), in a prospective 12-week open-label trial of 13 individuals on dialysis who had been denied transplant due to...

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high BMI, demonstrated an average weight reduction of 4.6 ± 2.4 kg with semaglutide. In a national cohort study of 151,649 individuals with type 2 diabetes initiating dialysis between 2013 and 2021, GLP-1–based therapy was associated with a 23% lower mortality and 66% higher chance of transplant waitlisting (157).

In an observational study, use of GLP-1–based therapy at the onset of dialysis was associated with a decreased risk of major cardiovascular events (adjusted HR 0.65, P ≤ 0.001) and all-cause mortality (adjusted HR 0.63, P ≤ 0.001) during a median follow-up of 1.4 years. However, this study had methodological issues, including the likely inclusion of individuals with AKI (158). As mentioned before, exenatide and lixisenatide should not be used in individuals on dialysis.

Referral to a Nephrologist

Recommendations

11.12a Individuals should be referred for evaluation by a nephrologist if they have rapidly increasing urinary albumin levels and/or rapidly decreasing eGFR and/or if the eGFR is ≤ 30 mL/min/1.73 m².

11.12b Refer to a nephrologist for uncertainty about the etiology of kidney disease or difficult management issues.

Health care professionals should consider referral to a nephrologist if the individual with diabetes has continuously rising UACR levels and/or continuously declining eGFR, if there is uncertainty about the etiology of kidney disease, for difficult management issues (anemia, secondary hyperparathyroidism, significant increases in albuminuria despite good blood pressure management, metabolic bone disease, resistant hypertension, or electrolyte disturbances), or when there is advanced kidney disease (eGFR $\leq$ 30 mL/min/1.73 m²) requiring discussion of kidney replacement therapy for kidney failure (1). The threshold for referral may vary depending on the frequency with which a health care professional encounters people with diabetes and kidney disease. Consultation with a nephrologist when stage G4 CKD develops (eGFR $\leq$ 30 mL/min/1.73 m²) has been found to reduce cost, improve quality of care, and delay the need for dialysis (159).

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